6 Mathematical Induction

The idea of induction

Your friend Pablo P. Casso is planning to assign colors to the positive integers

$$1, 2, 3, 4, 5, 6, 7, \ldots$$

You don't know what the assignment is going to be, but you know this:

- The integer 1 is going to be blue.
- Any integer next to a blue integer is going to be blue.

Then you know that *all* the integers are going to be blue:

$$1, 2, 3, 4, 5 \ldots$$

This is the "rule of induction" or "the induction principle" at work.

The math induction principle (summarized in the box on page 67)

More generally, the induction principle is the following. Assuming that you have some property of numbers, call it $P(n)$ (in our example, $P(n)$ is the property "integer n is colored in blue"), such that

- you know $P(1)$ to be true,
- and you also know that $P(1)$ implies $P(2)$, $P(2)$ implies $P(3)$, $P(3)$ implies $P(4)$, ..., $P(k)$ implies $P(k+1)$, ...,

then you can conclude that $P(n)$ is true for all $n \geq 1$.

The idea of induction is also often likened to the process of knocking over a row of dominoes. To ensure that all dominoes fall,

- You have to knock over the first domino. ($P(1)$ is true.)
- You have to be sure that when domino k falls, it knocks over domino $k+1$ for all $k \geq 1$. ($P(k)$ implies $P(k+1)$ for all $k \geq 1$.)

6.1 The Principle of Mathematical Induction

Induction can be used to prove that some property $P(n)$ holds for every number $n = 1, 2, 3$, and so on. A proof by induction has two parts: a *basis step*, where we show that $P(1)$ is true, and an *inductive step*, where we show that for any positive integer k, if $P(k)$ is true, then $P(k+1)$ is true.

The Principle of Mathematical Induction

Let n be some positive integer and let $P(n)$ be some statement about n that is either true or false.

To prove that $P(n)$ is true for all n, complete two steps:

Basis Step: Verify that $P(1)$ is true.

Inductive Step: Show that the conditional statement "$P(k)$ implies $P(k+1)$" is true for all positive integers k.

predicate: $P(n)$.

The statement about n – which we denoted here by $P(n)$ – is sometimes called the *predicate* (because it says something about the integer n) or the *inductive hypothesis* about n (because it hypothesizes that something is true about n). Examples of $P(n)$ are propositions like "the integer n is odd" or "the integer n is colored blue."

Thus, for a fixed n, $P(n)$ is a logical *P*roposition. As such, it is either true or false. Our job in inductive proofs is to show that $P(n)$ is true for any n in the specified range.

the basis step doesn't have to start at 1

The specified range for n in the box above is $n \geq 1$. But it doesn't have to be. If you are interested in proving that $P(n)$ is true for all $n \geq 3$, for example, then the *Basis Step* in the box above would read "Verify that $P(3)$ is true."

shorthand notation for "implies" is "→"

In what follows, we sometimes write "implies" as "→." For example, we may write "$P(1)$ implies $P(2)$" as "$P(1) \to P(2)$."

All proofs by mathematical induction follow a very predictable pattern – a template you fill in to adapt to your particular problem. That template is in Section 6.3 (on page 71). First, though, we'll look at an example.

6.2 An Example Proof by Mathematical Induction

to do:

1. *scratchwork*
2. *edit proof*

Example 23 is our first example of a proof by induction. We first want to show in detail the steps involved in proofs by induction. We then write a short, clean, edited version of the proof.

So *do* pay careful attention to the steps involved in this discussion. All inductive proofs follow the same steps, which we summarize in the box on page 71. You will be following the same steps every time you write an inductive proof on a test or for homework.

Example 23. Prove the following fact using the principle of mathematical induction.

Fact: For any integer $n \geq 1$,

$$1 + 2 + 3 + \ldots + (n-1) + n = \frac{n(n+1)}{2} \tag{6.1}$$

When writing down an inductive proof, the first thing to do is to say precisely what your predicate $P(n)$ is. Often, $P(n)$ is lifted directly from the proposition you are asked to prove, as in this example.

Step 1: write what $P(n)$ is

Define the predicate $P(n)$ to be equation (6.1):

$$P(n) \text{ is the proposition “ } 1 + 2 + \ldots + n = \frac{n(n+1)}{2}. \text{ ”}$$

We wish to prove that $P(n)$ is true for all $n \geq 1$. This is great because the induction principle lets us reach precisely that conclusion, provided we establish two simpler facts:

- ⋄ $P(1)$ is true. *(The Basis Step)*
- ⋄ For any $k \geq 1$, $P(k)$ implies $P(k+1)$. *(The Inductive Step)*

Step 2: show the basis step holds

The first statement follows because the left-hand side of equation (6.1) is equal to 1, and its right-hand side is also equal to $1(1+1)/2 = 1$.

- ⋄ So $P(1)$ is true. ☑

Step 3: show the inductive step holds

To verify the inductive step, we must show that $P(k) \to P(k+1)$. And to do that, we assume $P(k)$ to be true and show that then $P(k+1)$ must necessarily be true[1].

(step 3 often takes a bit more effort than the other steps)

So, assume that $P(k)$ is true. Namely, we assume

$$1 + 2 + \ldots + k = \frac{k(k+1)}{2} \text{ to be true} \tag{6.2}$$

in order to prove that $P(k+1)$ is true. Namely, to prove that

$$1 + 2 + \ldots + (k+1) = \frac{(k+1)(k+2)}{2} \text{ is true.} \tag{6.3}$$

Now note that adding $k+1$ to both sides of equation (6.2) and simplifying the right-hand side gives $P(k+1)$, equation (6.3):

$$\begin{aligned} 1 + 2 + 3 + \ldots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \qquad \text{(add } k+1 \text{ to (6.2))} \\ &= \frac{(k+1)(k+2)}{2}. \qquad \text{(simplify using algebra)} \end{aligned}$$

Thus, we conclude that whenever $P(k)$ is true, then so is $P(k+1)$:

- ⋄ So $P(k) \to P(k+1)$ is true for any $k \geq 1$. ☑

Step 4: conclude your proof

To conclude the proof, say that you've verified both the basis and the inductive steps of the induction principle and that, therefore, $P(n)$ has to be true for any $n \geq 1$. □

[1] Don't let this confuse you. We are *not* saying that $P(k)$ *is* true. (If we did, we'd be reasoning circularly: we'd be assuming exactly what we want to prove.)

Instead, we are saying "whenever $P(k)$ is true, then so is $P(k+1)$," which makes the implication $P(k) \to P(k+1)$ true.

Put another way, we are verifying that the implication $P(k) \to P(k+1)$ is true by verifying that "there is no way $P(k+1)$ is false if $P(k)$ is true." This works because the *only* time the implication is false is when $P(k)$ is true and $P(k+1)$ is false.

A Clean Write-up

The proof of Fact 6.1 we gave in Example 23 is perfectly valid; however, it contains a lot of extraneous explanations that you won't usually see in inductive proofs. The write-up below is closer to what you might see in print and should be prepared to produce yourself.

A revised proof of Fact 6.1 from Example 23.

Proof is by mathematical induction.

Let the inductive hypothesis $P(n)$ be equation (6.1).

Basis step. $P(1)$ is true, because both sides of equation (6.1) are equal to 1 when $n = 1$.

Inductive step. Assume that $P(k)$ is true. That is, assume equation (6.2) holds for some nonnegative integer k. Then adding $k + 1$ to both sides of the equation implies that

$$\begin{aligned} 1 + 2 + 3 + \ldots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}, \qquad \text{(by algebra)} \end{aligned}$$

which proves that $P(k + 1)$ also holds.

We have shown that both the basis step and the inductive step of the principle of mathematical induction hold.

Therefore, it follows from the principle of mathematical induction that $P(n)$ is true for all nonnegative n. This completes the proof.

It may bother you that induction led to a proof of the summation formula 6.1 but did not provide an intuitive way to understand it, nor did it explain where the formula came from in the first place. This is both a weakness and a strength. It is a weakness when a proof does not provide insight. But it is a strength that a proof can provide a reader with a reliable guarantee of correctness without *requiring* insight.

6.3 A Template for Inductive Proofs

Even the most involved proofs by induction proof follow exactly the same template. There are five components[2]:

1. **State that the proof uses induction.** This immediately conveys the overall structure of the proof, which helps your reader follow your argument.

2. **Define an appropriate predicate $P(n)$.** The predicate $P(n)$ is called the *inductive hypothesis*. The eventual conclusion of the inductive argument will be that $P(n)$ is true for all positive n. A clearly stated inductive hypothesis is often the most important part of an inductive proof, and its omission is the largest source of confused proofs by students.

   In the simplest cases, the induction hypothesis can be lifted straight from the proposition you are trying to prove, as we did with equation (6.1). Sometimes the induction hypothesis will involve several variables, in which case you should indicate which variable serves as n.

3. **Prove that $P(1)$ is true.** This is usually easy, as in the example above. This part of the proof is called the *base case* or *basis step*.

4. **Prove that $P(k)$ implies $P(k+1)$ for every positive integer k.** This is called the *inductive step*. The basic plan is always the same: assume that $P(k)$ is true, and then use this assumption to prove that $P(k+1)$ is true. These two statements should be fairly similar, but bridging the gap may require some ingenuity. Whatever argument you give must be valid for every positive integer k, since the goal is to prove that all the following implications are true:

   $$P(1) \text{ implies } P(2), \quad P(2) \text{ implies } P(3), \quad P(3) \text{ implies } P(4), \ldots$$

5. **Invoke induction.** Given these facts, the induction principle allows you to conclude that $P(n)$ is true for all positive n. This is the logical capstone to the whole argument, but it is so standard that it's usual not to mention it explicitly.

[2] When answering quiz questions about inductive proofs, you can usually omit step 1: you don't have to say in your answer that you are about to do a proof by induction when the question explicitly asks you to do just that.

6.4 Other Examples Completed in Class

Example 24. Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

(For example, when $n = 5$, $n^3 - n = 5^3 - 5 = 125 - 5 = 120$, which is divisible by 3: $120 \div 3 = 40$. Your job is to prove that this holds for every $n \geq 1$.)

Read Example 8 on page 343 of Rosen's textbook

We proved this in class on Monday, Feb 13. See your class notes and the textbook for a proof. Keep an eye on how the textbook's proof follows the proof template on page 71.

Example 25. Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.

(Here are the sums of the first few odd integers:

$$\begin{aligned} 1 &= 1 \\ 1 + 3 &= 4 \\ 1 + 3 + 5 &= 9 \\ 1 + 3 + 5 + 7 &= 16 \\ 1 + 3 + 5 + 7 + 9 &= 25 \\ 1 + 3 + 5 + 7 + 9 + 11 &= 36 \\ &\vdots \end{aligned}$$

Did you pick up on the pattern? Make a conjecture. Your job is to prove that your conjecture is true.)

I'll prove this in class – with your help – on Friday, Feb 17.

Example 26. Prove the following claim using mathematical induction.

Claim. For all positive integers n, we have

$$2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + (n+1) \cdot 2^n = n \cdot 2^{n+1}.$$

proof Proof is by mathematical induction. For every integer $n \geq 1$, let $P(n)$ be the proposition

$$2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + (n+1) \cdot 2^n = n \cdot 2^{n+1}. \tag{6.4}$$

Basis step. $P(1)$ is true because $2 \cdot 2 = 1 \cdot 2^{1+1}$.

Inductive step. Let k be any integer such that $k \geq 1$, and assume that $P(k)$ is true. That is, assume that

$$2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + (k+1) \cdot 2^k = k \cdot 2^{k+1}.$$

Our goal is to prove that $P(k+1)$ is also true. That is, we wish to show that

$$2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + ((k+1)+1) \cdot 2^{k+1} = (k+1) \cdot 2^{(k+1)+1} \quad \text{is true.}$$

In other words, simplifying the above equation a bit, we wish to prove that

$$2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + (k+2) \cdot 2^{k+1} = (k+1) \cdot 2^{k+2}. \tag{6.5}$$

We have:

$$\begin{aligned}
\text{Left-hand-side of (6.5)} &= \underbrace{2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \ldots + (k+1) \cdot 2^k} + (k+2) \cdot 2^{k+1} \\
&= \underbrace{k \cdot 2^{k+1}} + (k+2) \cdot 2^{k+1} \quad \text{(because } P(k) \text{ is true)} \\
&= (k + (k+2)) \cdot 2^{k+1} \\
&= (2k+2) \cdot 2^{k+1} \\
&= (k+1) \cdot 2 \cdot 2^{k+1} \\
&= (k+1) \cdot 2^{k+2} \\
&= \text{Rigt-hand-side of (6.5)}
\end{aligned}$$

Thus, we conclude that $P(k+1)$ is true as long as $P(k)$ is true. We just showed that $P(k)$ implies $P(k+1)$, for every $k \geq 1$, concluding the proof of the inductive step.

By the principle of mathematical induction, it follows that $P(n)$ is true, for all $n \geq 1$. This completes the proof. $\square$

Example 27. *(a bogus proof?)* Prove the following by induction

Theorem: All horses are the same color.

Suppose you begin your inductive proof by following the template on page 71. Part 2 of the template says that you need to "define the appropriate predicate $P(n)$". The statement of this theorem cannot be used as a predicate – there is no number n in it to induct on. But you can reformulate the theorem as

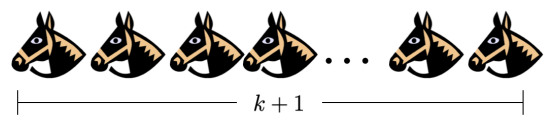
Theorem: In any set of $n \geq 1$ horses, all the horses are the same color.

proof The proof is by induction on n. The inductive hypothesis $P(n)$ will be

"In every set of n horses, all horses are of the same color."

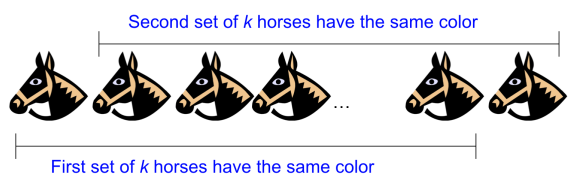
Basis step: $(k = 1)$. $P(1)$ is true, because in a size-1 set of horses, there's only one horse, and this horse is definitely the same color as itself.

Inductive step: Assume that $P(k)$ is true for some $k \geq 1$. That is, assume that in every set of k horses, all are the same color. Now suppose we have a set of $k + 1$ horses:



We need to prove these $k + 1$ horses are all the same color.

By our assumption, the first k horses are the same color. Also by our assumption, the last k horses are the same color.



Call these horses $h_1, h_2, \ldots, h_k, h_{k+1}$, from left to right. So h_1 is the same color as the remaining horses besides h_{k+1} – that is, $h_2, \ldots, h_k$. Likewise, h_{k+1} is the same color as the remaining horses besides h_1 – that is, $h_2, \ldots, h_k$ again. Since h_1 and h_{k+1} are the same color as $h_2, \ldots, h_k$, all $k + 1$ horses must be the same color, and so $P(k)$ implies $P(k + 1)$.

Thus, by the principle of induction, $P(n)$ is true for all $n \geq 1$. $\square$

6.5 Strong Induction

A useful variant of induction is called *strong induction*. Strong induction and ordinary induction are used for exactly the same thing: proving that a predicate is true for all positive integers. Strong induction is useful when a simple proof that the predicate holds for $k+1$ does not follow just from the fact that it holds at k, but from the fact that it holds for other values $\leq k$.

The Principle of Strong Mathematical Induction

Let n be some positive integer and let $P(n)$ be some statement about n that is either true or false.

To prove that $P(n)$ is true for all n, complete two steps:

Basis Step: Verify that $P(1)$ is true.

Inductive Step: Show that for all $k \geq 1$,

$$P(1), P(2), \ldots, P(k) \ \underline{\textit{together}} \text{ imply } P(k+1)$$

Compare this with the principle of math induction on page 67. The only change from the ordinary induction principle is that strong induction allows you make more assumptions in the inductive step of your proof! In an ordinary induction argument, you assume that $P(k)$ is true and try to prove that $P(k+1)$ is also true. In a strong induction argument, you may assume that $P(1), P(2), \ldots,$ and $P(k)$ are all true when you go to prove $P(k+1)$. So you can assume a stronger set of hypotheses which can make your job easier.

6.6 Examples of Proofs by Strong Induction

So, to prove the $(k+1)$-st step, the principle of strong induction allows us to apply not just the k-th step, but any of the steps $1, 2, \ldots, k$.

Example 28. Prove that every integer $n \geq 11$ can be written as $2a + 5b$ for some positive integers a and b.

☞ *Scratch work*. Before attempting a proof, always do some scratch work to figure out what you are asked to prove makes sense. In this example, you could try a few values of n :

$$\begin{array}{lll} 11 = 2 \cdot 3 + 5 \cdot 1 & 13 = 2 \cdot 4 + 5 \cdot 1 & 15 = 2 \cdot 5 + 5 \cdot 1 \\ 12 = 2 \cdot 1 + 5 \cdot 2 & 14 = 2 \cdot 2 + 5 \cdot 2 & 16 = 2 \cdot 3 + 5 \cdot 2 \\ \quad \cdots & & 2166 = 2 \cdot 3 + 5 \cdot 432 \end{array}$$

You may also notice here that the representation doesn't have to be unique. For example, $2166 = 2 \cdot 3 + 5 \cdot 432$ and $2166 = 2 \cdot 8 + 5 \cdot 430$ are two different ways to write the number 2166.

Catch an insight while doing scratch work?

When doing scratch work like this, you may often catch an insight or two. For example, you may notice that to move from 14 to 16, all you need is an extra 2. That is, $14 = 2 \cdot 2 + 5 \cdot 2$ and $16 = 2 \cdot 3 + 5 \cdot 2$. You can work this insight into the proof by realizing that you can write any $(k + 1)$ in this way by adding on an extra 2 if you get to assume that any $(k - 1)$ can be written this way. So, here's the proof.

Proof: The proof is by strong induction. For any $n \geq 11$, let $P(n)$ be the proposition that $n = 2a + 5b$ for some positive integers a and b.

<u>*Basis Step*</u>. The two propositions, $P(11)$ and $P(12)$, will constitute our basis step. Both of these propositions are true because

$$11 = 2 \cdot 3 + 5 \cdot 1 \qquad \text{and} \qquad 12 = 2 \cdot 1 + 5 \cdot 2.$$

<u>*Inductive Step*</u>. Fix any $k \geq 12$ and suppose that propositions $P(11)$ through $P(k)$ are all true. We want to show that $P(k + 1)$ is also true. Because, in particular, we can assume that $P(k - 1)$ is true, we have that

$$k - 1 = 2a + 5b \quad \text{for some positive integers } a \text{ and } b.$$

Adding 2 to both sides of the above, we get

$$k + 1 = 2(a + 1) + 5b,$$

where $a + 1$ and b are positive integers. This proves that $P(k + 1)$ is true, completing the inductive step.

We conclude that by the principle of strong induction, $P(n)$ is true for every $n \geq 11$. □

Example 29. Prove the following claim using the principle of strong induction.

Claim. Every integer greater than 1 is a product of primes.

proof Proof is by strong induction. Let the inductive hypothesis $P(n)$ be

$$n \text{ is a product of primes.}$$

Our goal is to show that $P(n)$ holds for all $n \geq 2$.

Basis Step: $(k = 2)$. $P(2)$ is true because 2 is prime, so it is a length one product of primes by convention.

Inductive step: Suppose that $k \geq 2$ and that every number from 2 to k is a product of primes. We must show that $P(k+1)$ holds, namely, that $k+1$ is also a product of primes. We argue by cases:

1. If $k+1$ is itself prime, then it is a length one product of primes by convention, and so $P(k+1)$ holds in this case.
2. Otherwise, $k+1$ is not prime, which by definition means $k+1 = a \cdot b$ for some integers a, b between 2 and k. Now by the strong induction hypothesis, we know that both a and b are products of primes. By multiplying these products, it follows immediately that $a \cdot b = k+1$ is also a product of primes. Therefore, $P(k+1)$ holds in this case as well.

Thus, $P(k+1)$ holds in either case.

We have proved that both the basis and the inductive steps of the strong mathematical induction principle hold.

So, according to the principle of strong mathematical induction, $P(n)$ holds for all $n \geq 2$. This completes the proof.

Example 30. Prove the following claim.

Claim. All natural numbers are interesting.

We prove this claim using strong induction. For each natural number n, let $P(n)$ be the sentence:

"The number n is interesting."

Basis step: $P(1)$ is 'The number 1 is interesting.' This is clearly true.

Inductive step: Let k be a natural number and assume that

$$P(1), P(2), \ldots, P(n)$$

are true, that is, assume that the numbers $1, 2, \ldots, k$ are interesting. We wish to use this to prove $P(k+1)$, i.e., that the number $k+1$ is interesting.

Suppose, by way of contradiction, that the number $k+1$ is not interesting. By assumption, the numbers $1, 2, \ldots, k$ are interesting, so it follows that $k+1$ is the *first uninteresting natural number*, and that makes it very interesting, which is a contradiction!

Since our assumption that $k+1$ was uninteresting led to a contradiction, this assumption is false, and hence $k+1$ is interesting. Thus, $P(k+1)$ is true.

By the principle of strong mathematical induction, it follows that $P(n)$ is true for every natural number n, i.e., *every natural number is interesting.* □

You thought the claim in Example 30 was strange

If a proof relies on beliefs, it may be departing from mathematics

What have we actually proved in Example 30? A reasonable interpretation is that, if we believe that 1 is an interesting natural number, then every natural number is interesting. The proof comes down to two statements of *belief* about what is, or is not, interesting:

- the natural number 1 is interesting;
- being the smallest uninteresting natural number would make a number interesting.

If you reject either of these claims, then the proof doesn't work. If you accept them, then you have to accept that all natural numbers are interesting.

Example 31. The country of Oz issues only 3-cent and 8-cent stamps. What amounts of postage are possible with just these two kinds of stamps?

scratchwork By experimentation, we can find out that all of

$$0,\ 3,\ 6,\ 8,\ 9,\ 11,\ 12,\ 14,\ 15,\ 16,\ 17,\ 18,\ 19,\ 20,\ \text{and } 21$$

cents are possible. Since we are getting all amounts of 14 cents or greater,we conjecture that all amounts except $1, 2, 4, 5, 7, 8, 10$, and 13 cents are possible.

Looks like we'll have to handle the possible and the impossible postage amounts separately. Let's start with what postage amounts are possible.

Conjecture: Any amount of postage of 14 cents or more can be formed using just 3-cent and 8-cent stamps.

proof We'll use the principle of strong mathematical induction to prove the conjecture. Let $P(n)$ be the inductive hypothesis that for any $n \geq 14$, $n = x \cdot 3 + y \cdot 8$ for some integer $x, y \geq 1$.

Basis step. After some experimentation, we decide the base cases are $14, 15$, and 16. Since $14 = 2 \cdot 3 + 1 \cdot 8, 15 = 5 \cdot 3 + 0 \cdot 8$, and $16 = 0 \cdot 3 + 2 \cdot 8$, we have that $P(14), P(15)$, and $P(16)$ are all true.

Inductive step. Let $k \geq 14$ and assume that $P(14), P(15), P(16), \ldots, P(k)$ are all true. We now prove that $P(k+1)$ is also true.

Since $14, 15$, and 16 are base cases, every possible value for $k+1$ that is not a base case and is greater than or equal to 14 is also greater than or equal to 17. For $k+1 \geq 17$, we have $(k+1) - 3 \geq 14$. So, by the inductive hypothesis, for some $x, y \geq 1$, $(k+1) - 3 = x \cdot 3 + y \cdot 8$. Then

$$k + 1 = (k+1) - 3 + 3 = x \cdot 3 + y \cdot 8 + 3 = (x+1) \cdot 3 + y \cdot 8$$

So, $P(k+1)$ is true, as desired.

By the strong form of mathematical induction, $P(n)$ is true for all $n \geq 14$.

What about the postage amounts $n < 14$? We noted in our scratchwork that 0, 3, 6, 8, 9, 11, and 12 cents are all possible.

But the amounts $1, 2, 4, 5, 7, 8, 10$, or 13 cents are impossible: To get any of these amounts, you would need to use, at most, 4 stamps (why?), and we can list all the possible combinations of $0 - 4$ stamps to show that none add up to $1, 2, 4, 5, 7, 8, 10$, or 13 cents.

6.7 Strong Induction vs. Ordinary Induction

Strong induction looks genuinely "stronger" than ordinary induction – after all, you can assume a lot more when proving the inductive step. Since ordinary induction is a special case of strong induction, you might wonder why anyone would bother with ordinary induction.

Anything that can be proved by strong induction can also be proved by ordinary induction using *essentially the same proof*! In fact, a simple text manipulation program with no built-in theorem-proving ability can mindlessly reformat any proof using strong induction into a proof using ordinary induction. This is done just by replacing the strong induction hypothesis $P(n)$ with the ordinary induction hypothesis $Q(n) ::= \forall k \leq n.\ P(k)$.

Still, it's worth distinguishing these two kinds of induction since which you use will signal whether the inductive step for $n + 1$ follows directly from the case for n or requires cases smaller than n, which is generally good for your reader to know. The point is that, theoretically, induction and strong induction are simply different formats for presenting the same mathematical reasoning!

6.8 Exercises

6.1. What is wrong with the proof in Example 27?

6.2. Prove these inequalities for the indicated values of n using mathematical induction.

(a) $3^n > 2n$ for $n = 1, 2, 3, \ldots$

(b) $n^2 > 2n + 1$ for $n = 3, 4, 5, \ldots$

(c) $2^n > n^2$ for $n = 5, 6, 7, \ldots$

(d) $2^n < n!$ for $n = 4, 5, 6, \ldots$

(e) $(1 + x)^n \geq 1 + nx$ for $n = 1, 2, 3, \ldots$

6.3. Prove that for any positive integer n the following identity holds

$$1 + 2^1 + 2^2 + 2^3 + \ldots + 2^n = 2^{n+1} - 1$$

6.4. Prove by induction:

$$1 + \frac{1}{4} + \frac{1}{9} + \ldots + \frac{1}{n^2} < 2 - \frac{1}{n},$$

for all $n > 1$.

6.5. Although the term *mathematical induction* was introduced and the process was put on a rigorous basis by the British mathematician Augustus De Morgan in 1838, informal induction-type arguments have been used as far back as the 10th century. The Persian mathematician al-Karaji (953 – 1029) essentially gave an induction-type proof of the formula for the sum of the first n cubes: $1^3 + 2^3 + \ldots + n^3 = (1 + 2 + \ldots + n)^2$.

It's now your turn. Prove by induction that

$$1^3+2^3+\ldots+n^3 = (1+2+\ldots+n)^2 = \left(\frac{n(n+1)}{2}\right)^2.$$

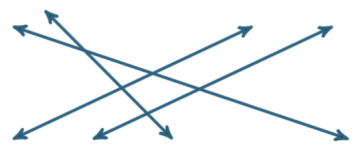
6.6. Prove that 6 divides $n^3 - n$ whenever n is a nonnegative integer.

6.7. Prove that $n^2 - 7n + 12$ is nonnegative whenever $n \geq 3$.

6.8. Prove that $n^2 - 1$ is divisible by 8 whenever n is an odd positive integer.

6.9. Prove that 21 divides $4^{n+1} + 5^{2n-1}$ whenever n is a positive integer.

6.10. The plane is divided into regions by straight lines. Show that it is always possible to color the regions with 2 colors so that adjacent regions are never the same color.



Hint: Use induction on the number of lines.

6.11. Lines in a plane are said to be in a general position if no 2 are parallel and no 3 meet in a point. If 10 lines are drawn in general position in the plane, into how many regions do they divide the plane?

6.12. Use mathematical induction to prove that if $n \geq 1$, then the following statements are true.

(a) $17^n - 12^n$ is divisible by 5.

(b) $n^3 + n$ is even (for this problem, try also to find a proof that does not use induction; it should be simpler than an inductive proof).

6.13. Use mathematical induction to prove that for $n > 23$, there exist nonnegative integers x and y such that $n = 7x + 5y$.

6.14. Use strong induction to show that every positive integer n can be written as a sum of distinct powers of two, that is, as a sum of a subset of the integers $2^0 = 1, 2^1 = 2, 2^2 = 4$, and so on.

(For example, $5 = 2^0 + 2^2$ and $2166 = 2^1 + 2^2 + 2^4 + 2^5 + 2^6 + 2^{11}$, etc.)

6.15. Find the flaw with the following "proof" that $a^n = 1$ for all nonnegative integers n, whenever a is a nonzero real number.

Basis step: $a^0 = 1$ is true by the definition of a^0.

Inductive Step: Assume that $a^j = 1$ for all nonnegative integers j with $j \leq k$. Then note that

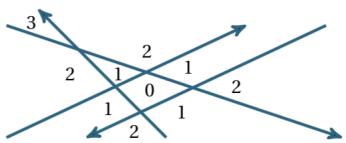
$$a^{k+1} = \frac{a^k \cdot a^k}{a^{k-1}} = \frac{1 \cdot 1}{1} = 1.$$

Answers to Selected Problems

6.1 It never proves that $P(1)$ implies $P(2)$. This is a subtle, hard-to-detect bug. But some inductive proofs, even those published by seasoned mathematicians, can suffer from subtle flaws like this.

6.10 We give a non-inductive proof here. See if you can follow the usual inductive proof template (page 71) and write down an inductive proof.

Proof (non-inductive). Assume that we have a finite number of straight lines in the plane. Fix a direction on each line in any way at all. (An example is given by the arrowheads in the following diagram.) Now it makes sense to talk about the left side and the right side of each line. For each region R, count the number n_R of lines ℓ such that the region R is on the left side of ℓ. (The number n_R is given in each region R in the following diagram.)



Now color a region R white if n_R is even, and black if n_R is odd.

Assume that R and S are regions that share a border that is part of the line ℓ. We must prove that R and S have different colors. Since R and S share a border that is part of ℓ, one of the regions is to the left of ℓ and the other is to the right of ℓ. Moreover, the regions R and S are on the same side of every other line. Thus, the numbers n_R and n_S must differ by 1, and so one of the numbers is even and the other is odd. Consequently, the regions R and S have different colors, as required.

6.13 This is very similar to Example 31. The following is essentially the same proof as the one given in that example, but it asks you to fill in the details. Read the example again if this proof is not clear after you carefully read it.

Proof. We argue by cases, proving each case with induction.

We have 5 cases: n could have remainder $0, 1, 2, 3,$ or 4 upon division by 5.

Establishing the basis step for cases involving remainders of $4, 0, 1, 2,$ and 3 (in that order), we observe:

- $24 = 7 \cdot 2 + 5 \cdot 2$
- $25 = 7 \cdot 0 + 5 \cdot 5$
- $26 = 7 \cdot 3 + 5 \cdot 1$
- $27 = 7 \cdot 1 + 5 \cdot 4$
- $28 = 7 \cdot 4 + 5 \cdot 0$

To argue the inductive step for each case, notice that if the statement is true for some $n = k$, then $k = 7x + 5y$, which in turn implies $k + 5 = 7x + 5(y + 1)$. As $n = k + 5$ is the next value of n with the same remainder as k, the inductive step is established.

Therefore, by the principle of mathematical induction, for $n > 23$, there must always exist nonnegative integers x and y such that $n = 7x + 5y$.

6.14 We use strong induction to prove the claim $P(n)$:

"Every $n \geq 1$ can be represented as a sum of distinct powers of 2."

Basis step: Because $1 = 2^0$, $P(1)$ is true.

Inductive step: We now show that $P(1), P(2), \ldots, P(k)$ *together* imply $P(k + 1)$ for any $k \geq 1$. So we assume that *every* positive integer up to k can be represented as a sum of distinct powers of 2, and consider the integer $k+1$. There are two cases:

Case 1: $k + 1$ is odd; *Case 2:* $k + 1$ is even.

<u>*Case 1:*</u> If $k + 1$ is odd, then k is even, which means that the term 2^0 is not present in the sum that represents the integer k. But if we now add $2^0 = 1$ to the sum of the powers of 2 that represents k, we will have in our hands the representation of $k + 1$ as a sum of the powers of 2.

<u>*Case 2:*</u> If $k + 1$ is even, then $(k + 1)/2$ is a positive integer smaller than $k + 1$, and by our inductive hypothesis $(k+1)/2$ can be written as a sum of distinct powers of 2. If in this representation of $(k + 1)/2$ we increase the exponent of each term by 1 — which amounts to multiplying $(k + 1)/2$ by 2 — we will have in our hands the representation of k as a sum of distinct powers of 2.

This completes the inductive step.

By the principle of (strong) mathematical induction, we conclude that every positive integer can be represented as a sum of distinct powers of 2.

7 Induction & Recursive Definitions

explicitly defined f

One way to define a function is to specify its values by giving an explicit formula. For example, the formula $f(n) = 2^n$ for $n = 0, 1, 2, 3, \ldots$ defines a function named f over the domain of nonnegative integers; the values of f are given explicitly – those values are the powers of 2.

recursively defined f

But your could also define a function recursively. Recursive definitions involve two steps, one of which is a rule for generating "new" values of f from the "previous" values of f.

For example, to define the powers of 2 recursively, you proceed in two steps:

Step 1 is known as the basis step

Step 1: Specify the value of f at $n = 0$.

In our example, this step is

$$f(0) = 1.$$

Step 2 is known as the recursive step

Step 2: Specify a rule for finding $f(n)$ from the values of f at smaller n's.

In our example, this step is

$$f(n) = 2 \cdot f(n-1).$$

This rule tells us that if we knew the value of f at $n-1$, we would find $f(n)$ by multiplying that value by 2.

Notice how the powers of 2 are generated from this recursive definition

n	$f(n)$	Why?
0	1	Because Step 1 says so.
1	2	Because Step 2 says $f(1) = 2 \cdot f(0) = 2 \cdot 1$.
2	4	Because Step 2 says $f(2) = 2 \cdot f(1) = 2 \cdot 2$.
3	8	Because Step 2 says $f(3) = 2 \cdot f(2) = 2 \cdot 4$.
...	...	...

7.1 Recursively Defined Functions

Generally, recursive definitions follow this outline:

How to Define a Function Recursively

To define a function over the set of nonnegative integers, take two steps:

Basis Step: Give the value of the function at zero.

Recursive Step: Give a rule for finding the value of the function at any integer from its values at smaller integers.

Example 32. Consider the following recursive definition of f:

Basis Step: $f(0) = 0$.

Recursive Step: $f(n) = f(n-1) + 2$ when $n \geq 1$.

In all examples, f's domain is the set of nonnegative integers

This definition allows us to find the value of f at any nonnegative integer. Here are the first few values of f:

$$\begin{aligned}
f(0) &= 0 && \text{(by the basis step)}\\
f(1) &= 2 && \text{(by the recursive step, } f(1) = f(0) + 2)\\
f(2) &= 4 && \text{(by the recursive step, } f(2) = f(1) + 2)\\
f(3) &= 6 && \text{(by the recursive step, } f(3) = f(2) + 2)\\
f(4) &= 8 && \text{(by the recursive step, } f(4) = f(3) + 2)\\
&\cdots
\end{aligned}$$

Our next example, Example 33, shows how to define n-factorial recursively. (You are familiar with the conventional notation for the n-factorial function: most people write $n!$ and say "n-factorial.")

Here's the familiar definition of n-factorial that uses an explicit formula:

$$n! = 1 \cdot 2 \cdots (n-1) \cdot n \qquad \text{(explicit definition of } n)$$

Example 33 defines the exact same mathematical object – the $n!$ function – but does it recursively, not by specifying an explicit formula.

Example 33. Consider the following recursive definition of f:

Basis Step: $f(0) = 1$.

Recursive Step: $f(n) = n \cdot f(n-1)$ when $n \geq 1$.

Based on this definition, we can find the value of f at any nonnegative integer:

$$\begin{aligned}
f(0) &= 1 && \text{(by the basis step)}\\
f(1) &= 1 && \text{(by the recursive step, } f(1) = 1 \cdot f(0))\\
f(2) &= 2 && \text{(by the recursive step, } f(2) = 2 \cdot f(1))\\
f(3) &= 6 && \text{(by the recursive step, } f(3) = 3 \cdot f(2))\\
f(4) &= 24 && \text{(by the recursive step, } f(4) = 4 \cdot f(3))\\
&\ldots
\end{aligned}$$

Example 34. Consider the following recursive definition of f:

Basis Step: $f(0) = 0$ and $f(1) = 1$.

Recursive Step: $f(n) = f(n-1) + f(n-2)$ when $n \geq 2$.

Notice: the basis step can specify more than one initial value of f, and the recursive step can involve several earlier values of f.

This example differs slightly from the previous ones. First, the basis step specifies the values of f at two initial integers (not just one), at 0 and 1. Second, the recursive rule tells us that to find the value of f at any other integer, we need to know the values of f at two (not just one) integers immediately preceding it.

$$\begin{aligned}
f(0) &= 0 && \text{(by the basis step)}\\
f(1) &= 1 && \text{(by the basis step)}\\
f(2) &= 1 && \text{(by the recursive step, } f(2) = f(1) + f(0) = 1 + 0)\\
f(3) &= 2 && \text{(by the recursive step, } f(3) = f(2) + f(1) = 1 + 1)\\
f(4) &= 3 && \text{(by the recursive step, } f(4) = f(3) + f(2) = 2 + 1)\\
f(5) &= 5 && \text{(by the recursive step, } f(5) = f(4) + f(3) = 3 + 2)\\
f(6) &= 8 && \text{(by the recursive step, } f(6) = f(5) + f(4) = 5 + 3)\\
&\ldots
\end{aligned}$$

You may have recognized this famous sequence. If not, Exercise 7.2 will tell you the answer.

7.2 Induction & Recursion

Example 36 in the next chapter relates a recursive definition with simple induction.

You may have noticed how closely recursive definitions mimic the now-familiar principle of mathematical induction. It may come as no great surprise, then, that mathematical induction can be used to prove facts about recursively defined functions.

Example 35. Suppose the sequence $f(n)$ is defined over the set of positive integers recursively as follows:

Basis Step: $f(1) = 1,\ f(2) = 2,$ and $f(3) = 6.$

Recursive Step: $f(n) = (n^3 - 3n^2 + 2n)f(n-3)$ for all $n \geq 4$.

Then prove the following

<u>*Fact:*</u> If f is defined as above, then for all $n \geq 1,\ f(n) = n!$.

Does the question make sense? Do scratch work to see.

When faced with a question like this one, first do some scratch work to make sure what you are asked makes sense. Write down a few values of $f(n)$. Sure, the first three values generated by the basis step are the needed factorials:

$$f(1) = 1 = 1! \quad f(2) = 2 = 2 \cdot 1 = 2! \quad f(3) = 6 = 3 \cdot 2 \cdot 1 = 3!$$

How about a few other values generated by the recursive rule? Indeed,

$$f(4) = (4^3 - 3 \cdot 4^2 + 2 \cdot 4) \cdot f(1) = (64 - 48 + 8) \cdot 1 = 24 = 4!$$
$$f(5) = (5^3 - 3 \cdot 5^2 + 2 \cdot 5) \cdot f(2) = (125 - 75 + 10) \cdot 2 = 120 = 5!$$

These values work out. Does the pattern continue? Is it really true that Example 35 shows yet another way of defining the factorial function?

You can answer this with the help of mathematical induction.

We never had a chance to do this example in class, but some of you worked on it and wanted to see the answer. So I'm including the following proof for those who want to see it; since we never discussed this example, it will not appear on any of our tests.

Proof is by strong induction. For any $n \geq 1$, let $P(n)$ be the proposition $f(n) = n!$, where f is the recursively defined function of Example 35.

Basis Step:[1] We want to verify that $P(1), P(2)$, and $P(3)$ are true propositions. The truths of these three statements follow from the basis step of the recursive definition:

$$\begin{aligned} &P(1) \text{ is true because } f(1) = 1 = 1! \\ &P(2) \text{ is true because } f(2) = 2 = 2! \\ &P(3) \text{ is true because } f(1) = 6 = 3! \end{aligned}$$

Inductive Step:[2] Fix any $k \geq 4$ and suppose that $P(1), P(2), \ldots P(k)$ are all true. The goal is to prove that $P(k+1)$ must then also be true.

In other words, we want to show that if $f(i) = i!$ for all $i = 1, 2, \ldots k$, then $f(k+1)$ equals $(k+1)!$. We have:

$$\begin{aligned} f(k+1) &= [\,(k+1)^3 - 3(k+1)^2 + 2(k+1)\,] \cdot f(k-2) && \text{(by recursive definition)} \\ &= [\,(k+1)^3 - 3(k+1)^2 + 2(k+1)\,] \cdot (k-2)! && \text{(by inductive hypothesis)} \\ &= [\,k^3 - k\,] \cdot (k-2)! && \text{(by algebra)} \\ &= [\,(k+1)\,k\,(k-1)\,] \cdot (k-2)! && \text{(by algebra)} \\ &= (k+1)! \end{aligned}$$

This completes the inductive step.

So, by the principle of strong mathematical induction, $P(n)$ is a true proposition for any $n \geq 1$. □

[1] Note that this is the basis step of the principle of mathematical induction, not to be confused with the basis step of recursive definitions.

[2] Note that this is the inductive step of the principle of mathematical induction, not to be confused with the recursive step of recursive definitions.

7.3 Exercises

7.1. On page 84 we saw how to define the function $f(n) = 2^n$ recursively. Give a recursive definition of $f(n) = a^n$, where a is any nonzero real number and n is a nonnegative integer.

7.2. What is the name of the sequence generated by the definition in Example 34? (*Hint*: Google or Chat-GPT should give you the answer.)

7.3. Find $f(2), f(3), f(4)$ and $f(5)$, if f is defined recursively by $f(0) = -1$, and $f(1) = 2$, and for $n \geq 2$ as $f(n+1) = f(n)^2 f(n-1)$.

7.4. Find $f(2), f(3)$, and $f(4)$, if f is defined recursively by $f(0) = -1$, and $f(1) = 2$, and for $n \geq 2$ as $f(n+1) = \dfrac{f(n-1)}{f(n)}$.

7.5. Give a recursive definition of the function $f(n)$ over the domain of positive integers,$n \geq 1$, if

(a) $f(n) = 4n - 2$.
(b) $f(n) = 1 + (-1)^n$.
(c) $f(n) = n(n+1)$.
(d) $f(n) = n^2$.
(e) $f(n) = 6n$.
(f) $f(n) = 2n + 1$.
(g) $f(n) = 10^n$.
(h) $f(n) = 5$.

7.6. Let $a_0 = 1, a_1 = 3, a_2 = 9$, and define

$$a_n = a_{n-1} + 3a_{n-2} + 3a_{n-3}$$

for $n \geq 3$. Show by induction that $a_n \leq 3^n$ for all $n \geq 0$.

7.7. Define the sequence of numbers

$$a_n = \begin{cases} 1 & \text{if } 0 \leq n \leq 3, \\ a_{n-1} + a_{n-2} + a_{n-3} + a_{n-4} & \text{if } n \geq 4. \end{cases}$$

Prove that $a_n \equiv 1 \pmod 3$ for all $n \geq 0$.

7.8. Let the Tribonacci sequence be defined by $T_1 = T_2 = T_3 = 1$ and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$ for $n \geq 4$.

(a) Write down the first 10 elements of the Tribonacci sequence.
(b) Prove that $T_n < 2^n$ when n is a positive integer.

Answers to Selected Problems

7.1 *Basis Step:* Let $a^0 = 1$. *Recursive Step:* The rule for finding a^{n+1} from a^n is $a^{n+1} = a \cdot a^n$. These two steps uniquely define a^n for all $n = 0, 1, 2, \ldots$.

7.3 The first few values generated by the recursive rule:

$$f(2) = 2^2 \cdot (-1) = -4$$
$$f(3) = (-4)^2 \cdot 2 = 32$$
$$f(4) = 32^2 \cdot (-4) = -4096$$
$$f(5) = (-4096)^2 \cdot 32 = 536,870,912$$

7.5 We give the answers to the parts (e) through (h):

(e) The sequence $f(n) = 6n$ is $1, 6, 12, \ldots$, that is, every 6-th integer. So the following recursive definition works: *Basis Step:* $f(1) = 6$. *Recursive Step:* $f(n) = 6 + f(n-1)$ whenever $n \geq 2$.

(f) The sequence $f(n) = 2n + 1$ is the sequence of odd numbers, beginning from 3. So the following recursive definition works: *Basis Step:* $f(1) = 3$. *Recursive Step:* $f(n) = f(n-1) + 2$ whenever $n \geq 2$.

(g) In the sequence $f(n) = 10^n$, each term is 10 times bigger than the one preceding it. So the following recursive definition works: *Basis Step:* $f(1) = 10$. *Recursive Step:* $f(n) = 10 \cdot f(n-1)$ whenever $n \geq 2$.

(h) The sequence $f(n) = 5$ is constant. We define it recursively as follows: *Basis Step:* $f(1) = 5$. *Recursive Step:* $f(n) = f(n-1)$ whenever $n \geq 2$.

7.7 Proof is by strong induction. For $n \geq 0$, let $P(n)$ be the proposition that $a_n \equiv 1 \,(\text{mod}3)$.

<u>*Basis Step:*</u> For $0 \leq k \leq 3$, $a_k = 1$ and is therefore $\equiv 1 \,(\text{mod}3)$. So $P(k)$ holds when $0 \leq k \leq 3$.

<u>*Inductive Step:*</u> For $k \geq 4$, assume $P(k)$ is true for all $0 \leq k \leq n$. We must show that $P(k+1)$ is then also true. Since each of $a_{k-4}, a_{k-3}, a_{k-2}$, and a_{k-1} is $\equiv 1 \,(\text{mod}3)$, their sum is $\equiv 4 \equiv 1 \,(\text{mod}3)$. So $a_k \equiv 1 \,(\text{mod}3)$ and $P(k+1)$ holds, concluding the inductive step.

By the principle of strong induction, $P(n)$ is true for all $n \geq 0$.

7.8 (a) 1,1,1,3,5,9,17,31,57,105, and so on. Every element beginning from the fourth is the sum of the three elements immediately preceding it. (b) Proof is by strong induction. For all $n \geq 1$, let $P(n)$ be the proposition

$$T_n < 2^n \tag{7.1}$$

<u>*Basis Step:*</u> For $k = 1, 2$, and 3, the proposition $P(k)$ is true:

$$k = 1\colon P(1) \text{ is true: } T_1 = 1 < 2^1$$
$$k = 2\colon P(2) \text{ is true: } T_2 = 1 < 2^2$$
$$k = 3\colon P(3) \text{ is true: } T_3 = 1 < 2^3$$

<u>*Inductive Step:*</u> Fix any $k \geq 4$ and suppose Proposition 7.1 holds true for all $n = 1, 2, \ldots, k$. Then

$$\begin{aligned} T_{k+1} &= T_k + T_{k-1} + T_{k-2} && \text{(by the definition of } a_n\text{)} \\ &< 2^k + 2^{k-1} + 2^{k-2} && \text{(by the inductive hypothesis)} \\ &= 2^{k+1}\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8}\right) \\ &= 2^{k+1}\,\frac{7}{8} \\ &< 2^{k+1} \end{aligned}$$

This completes the inductive step. By the principle of strong induction, proposition 7.1 is true for any $n \geq 1$.

8 Recursion & Structural Induction

We've brought up recursively defined objects several times in this course already. For example, in one of our earlier lectures, we saw a recursive definition of the factorial function (same as the Recursive Definition 8.1).

Here's the plan for the February 23 and February 26 lectures. We want to

- see an example of how math induction can be used to prove program correctness. In Section 8.1, we'll see why the factorial procedure of Snippet 8.1 always returns the correct value;
- see a few more examples of recursively defined objects;
- see how a variant of mathematical induction called structural induction can be used to prove useful truths about those objects.

8.1 Proving Program Correctness Using Mathematical Induction

You can use mathematical induction to prove the correctness of a recursive algorithm[1]. (By "correctness" we mean that the algorithm always returns the correct output.) Example 36 shows why the recursive procedure in Algorithm 8.1 is correct.

procedure *fact*

```
1 procedure fact(n : integer)
2   if n = 1 then return 1
3   return n * fact(n − 1)
```

Algorithm 8.1: calculates $n!$

Definition of $n!$

Define $n!$ for all $n \geq 1$ as follows:

Basis Step: Let $1! = 1$.

Recursive Step: For any other n, let $n! = n \cdot (n-1)!$

Recursive Definition 8.1: defines $n!$

Example 36. Prove that the procedure in Algorithm 8.1 returns the correct value of $n!$ (The values of $n!$ for $n \geq 1$ are defined in Recursive Definition 8.1).

proof Proof is by induction. For a positive integer n,

let $P(n)$ be the proposition that `fact(`n`)` of Algorithm 8.1 returns $n!$

Basis Step: $P(1)$ is true because `fact(`n`)` returns 1 immediately (line 2), and $1! = 1$ by definition.

Compare: the inductive step (of the math induction principle) vs. the recursive step (of the recursive definition)

Inductive Step: Suppose $P(k)$ is true for an arbitrary $k \geq 1$; namely, assume that `fact(`k`)` returns $k!$. We wish to prove that then $P(k+1)$ is also true; namely, that `fact(`$k+1$`)` returns $(k+1)!$. Indeed:

$$\begin{aligned} \texttt{fact}(k+1) &= (k+1) \cdot \texttt{fact}(k) && \text{(by line 3 in Alg. 8.1)} \\ &= (k+1) \cdot k! && \text{(by the inductive hypothesis)} \\ &= (k+1)! && \text{(by the definition of !)} \end{aligned}$$

notice how the inductive proof works in tandem with the recursive definition

We have shown that both the basis and the inductive steps of the principle of mathematical induction hold. Therefore, $P(n)$ is true for any $n \geq 1$. □

[1] You can also prove the correctness of *iterative* algorithms using induction. The basic idea in those proofs is to identify the algorithm's so-called *loop invariant*, which is a statement about the behavior of the algorithm's loop. Induction is then used to prove that the invariant starts out true and stays true throughout the execution of the algorithm. We mentioned this earlier when we justified the correctness of our sorting algorithms, but we will not pursue this further; you'll learn more about this in a course on algorithms.

8.2 Recursively Defined Sets

Example 37 shows how a set can be defined recursively. This is the same as example 5 on page 370 of your textbook.

Example 37. Consider the subset S of the set of integers recursively defined by

Basis step: $3 \in S$.

Recursive step: If $x \in S$ and $y \in S$, then $x + y \in S$

When we did this example in class, many of you conjectured that the set S given by this recursive definition consists of all multiples of 3. You did this by writing down a few elements generated by the definition – the basis step generates the number 3, the first application of the recursive step generates the number $3 + 3 = 6$, the next application of the recursive step generates the new numbers $3 + 6 = 9$ and $6 + 6 = 12$ and so on. □

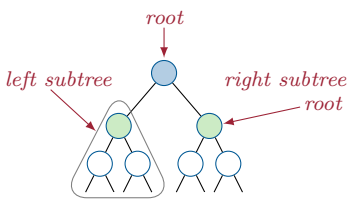
We then proved that the conjecture you made about the set S in Example 37 consists of all multiples of 3. That is, we proved the claim in Example 38.

Example 38. *Claim:* The set S defined in Example 37 consists of all multiples of 3.

Proof: Please see Example 10 on page 374 of your textbook and compare that proof with the notes you took in class. □

8.3 Trees

We'll give recursive definitions of ◇ binary trees and ◇ full binary trees.



The picture on the left shows the key terms associated with binary trees. The circles in the picture are called nodes or vertices and the lines connecting them are called edges.

One way to think of trees is to regard them as special kinds of graphs (loosely put, a graph is a collection of points and lines connecting some pairs of those points; we'll cover graphs later). But we can also define trees recursively.

Definition of a Binary Tree

- *Basis Step:* An empty object, denoted by $\emptyset$ is a binary tree.
- *Recursive Step:* If T_1 and T_2 are disjoint binary trees, then there is a binary tree T denoted by $T_1 \cdot T_2$ and consisting of a new node r called *root* together with edges connecting r to the roots of T_1 and T_2.

  T_1 is called the left subtree, and T_2 is called the right subtree of T.

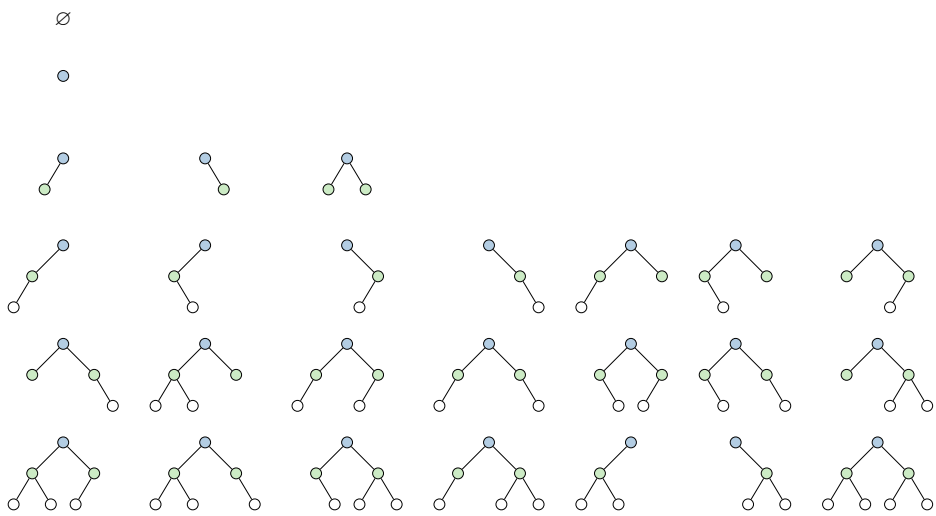
The following are the trees generated at the basis step and the first three recursive steps of the definition.

Basis Step:

Recursive Step 1:

Recursive Step 2:

Recursive Step 3:



The blue dots ● are the roots of the trees generated at each step and are connected to their subtree roots, marked by the green dots ●. Other tree nodes are marked by ○.

The definition of a full binary tree differs from the definition of a rooted tree given on the previous page only in the basis step.

Definition of a Full Binary Tree (FBT)

- *Basis Step:* A special node r called *root* root is a full binary tree.
- *Recursive Step:* If T_1 and T_2 are disjoint binary trees, then there is a binary tree T denoted by $T_1 \cdot T_2$ and consisting of a new node r called *root* together with edges connecting r to the roots of T_1 and T_2.

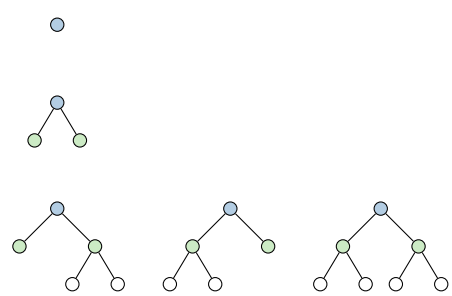
  T_1 is called the left subtree, and T_2 is called the right subtree of T.

These are the trees generated at the basis step and the first two recursive steps of the full binary tree definition:

Basis Step:

Recursive Step 1:

Recursive Step 2:



Notice that the trees generated by the recursive *full* binary tree definition are such that each node has either two or no children – no node can ever have a single child. By contrast, trees generated by the binary tree definition (on page 94) can have nodes with single children.

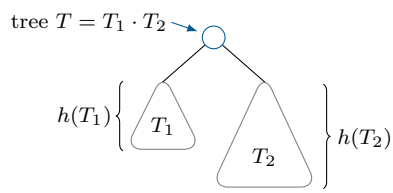
- ◇ tree height, denoted by $h(T)$;
- ◇ the number of nodes in a tree, denoted by $n(T)$;
- ◇ the number of leaves in a tree, denoted by $l(T)$;
- ◇ the number of internal nodes in a tree, denoted by $i(T)$.

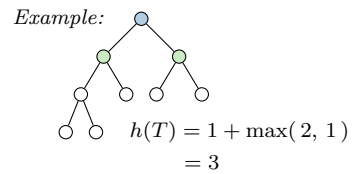
The definitions of these quantities are recursive.

Tree Height

Basis Step: The height of the full binary tree T consisting of only a root r is $h(T) = 0$.

Recursive Step: If T_1 and T_2 are disjoint full binary trees, then the full binary tree $T = T_1 \cdot T_2$ has height $h(T) = 1 + \max(\, h(T_1),\, h(T_2)\,)$.





Number of Tree Nodes

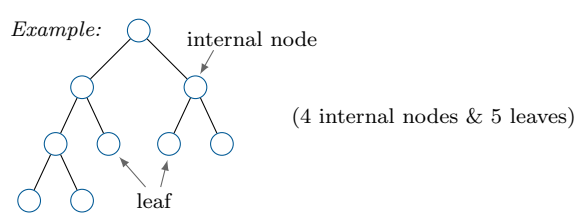
Basis Step: The number of nodes in the full binary tree T consisting of only a root r is $n(T) = 1$.

Recursive Step: If T_1 and T_2 are disjoint full binary trees, then the number of nodes in the full binary tree $T = T_1 \cdot T_2$ is $n(T) = 1 + n(T_1) + n(T_2)$.

Leaves and Internal Nodes

Basis Step: If a full binary tree T consists of only one root, r, then r is a leaf, and T has no internal nodes.

Recursive Step: If T_1 and T_2 are disjoint full binary trees, then the set of leaves of the full binary tree $T = T_1 \cdot T_2$ consists of the union of the set of leaves of T_1 and T_2. The internal nodes of T are the root r of T and the union of the set of internal nodes of T_1 and the set of internal nodes of T_2.



These quantities are related to one another, and to prove relationships among

them, we often involve a flavor of mathematical induction known as structural induction.

What Is Structural Induction?

Structural induction is a form of mathematical induction. It is useful in proving results about recursively defined objects (like binary trees). Proofs by structural induction consist of these two steps:

Basis Step: Show that the result holds for all the elements generated by the basis step of the recursive definition.

Recursive Step: Prove that if the result holds for all the elements used to generate new elements in the recursive step of the recursive definition, then the result also holds for those new elements.

Examples 39 and 40 relate the height of a full binary tree with the number of nodes in it. Loosely put, Example 39 says that a full binary tree T whose height is $h(T)$ "can't have too few nodes" and Example 40 says that it "can't have too many nodes."

As usual, before attempting a proof, make sure what you're asked makes sense to you. Draw a few full binary trees and relate the number of nodes in each tree to its height. Once you gain some intuition, try a proof. Because these quantities (the height $h(T)$ and the number of nodes $n(T)$) are defined recursively, a proof by structural induction makes sense.

Example 39. Show that $n(T) \geq 2h(T)+1$, where T is a full binary tree, $n(T)$ is the number of nodes in T, and $h(T)$ is the height of T.

Proof: Proof is by structural induction. We need to show that for any tree T generated by the recursive definition of a full binary tree (on page 95), the relationship $n(T) \geq 2h(T) + 1$ holds. (The definitions of $h(T)$ and $n(T)$ are on page 96.)

Basis Step: Suppose T is a full binary tree generated by the basis step of the definition (on page 95), so T consists of a single node, the root. According to the definitions of $n(T)$ and $h(T)$ on page 96, we have

$$1 = n(T) \geq 2h(T) + 1 = 2 \cdot 0 + 1 = 1.$$

Recursive Step: Now suppose that $T = T_1 \cdot T_2$ is generated at some recursive step of the definition on page 95. By the inductive hypothesis (IH), we have:

$$\textit{IH:} \qquad n(T_1) \geq 2h(T_1) + 1 \quad \text{and} \quad n(T_2) \geq 2h(T_2) + 1.$$

Using the recursive definitions of $n(T)$ and $h(T)$ on page 96, we have:

$$\begin{aligned}
n(T) &= 1 + n(T_1) + n(T_2) && \text{(by definition of } n(T)\text{)} \\
&\geq 1 + 2h(T_1) + 1 + 2h(T_2) + 1 && \text{(by the IH)} \\
&\geq 1 + 2\max(h(T_1), h(T_2)) + 2 && \text{(by arithmetic)} \\
&= 1 + 2(\max(h(T_1), h(T_2)) + 1) && \text{(by arithmetic)} \\
&= 1 + 2h(T). && \text{(by definition of } h(T)\text{)}
\end{aligned}$$

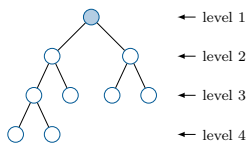
This completes the recursive step.

Therefore, by the principle of structural induction, the number of nodes $n(T)$ in any full binary tree T is at least as large as $2h(T) + 1$, where $h(T)$ is the height of T. □

Example 40. *Claim:* If T is a full binary tree, then $n(T) \leq 2^{h(T)+1} - 1$.

Proof: We did this proof in class in detail, so we will not repeat it here – what we did is very similar to what's in the textbook. So please see Theorem 2 on page 377 of your textbook and compare that proof with the notes you took in class. □

8.4 Exercises



8.1. The *level* of a node in a tree is a measure of its distance from the root. The root is considered to be at level 1, the root's children are at level 2, and so on. If you travel along the edges from the root to a node on any given level λ, you will visit λ nodes (including that node and the root). Define the level of a node recursively.

8.2. Let T be a binary tree with λ levels and n nodes, ℓ of which are leaves.

(a) Show that T has at most $2^{\lambda-1}$ leaves.

(b) Show that for every $k \geq 0$, there are no more than 2^k nodes at level k.

(c) Show that T has no more than $2^{\lambda-1}$ nodes.

(d) Show that the number of levels in T is at least $\lceil \log(n+1) \rceil$.

(e) Show that the number of levels in T is at least $\lceil \log \ell \rceil + 1$.

Exercise 8.3 says that in a full binary tree T, the total number of its nodes, n, the number of leaves ℓ, and the number of internal nodes i are related in a way that if you know any one of them, you can determine the other two.

8.3. Let T be a full binary tree (whose definition is on page 95). The notation we use in this exercise is as follows:

- ◇ ℓ: the number of leaves in T.
- ◇ i: the number of internal nodes in T.
- ◇ n: the total number of nodes in T (so $n = i + \ell$).

Prove the following facts:

(a) If T has i internal nodes, then it has $\ell = i + 1$ leaves.

(b) If T has i internal nodes, then it has $n = 2i + 1$ nodes.

(c) If T has n nodes, then it has $i = (n-1)/2$ internal nodes.

(d) If T has n nodes, then it has $\ell = (n+1)/2$ leaves.

(e) If T has ℓ leaves, then it has $n = 2\ell - 1$ nodes.

(f) If T has ℓ leaves, then it has $i = \ell - 1$ internal nodes.

Suggestion: Use induction. Draw a few examples before writing down your proof. For example, for part (a) you could draw a small tree like the one shown in the definition of leaves and internal nodes at the bottom of page 96.

Answers or Hints to Selected Problems

8.1 A recursive definition of the node level:

Basis Step:	Root level is 1.
Recursive Step:	Node level is $1 +$ the level of its parent.

8.2 We just give an inductive proof outline for part (a). Other parts are proved similarly by induction. The *basis step* generates tree T that only has $\lambda = 1$ level – there is only one root node in T and that node is a leaf node. So the claim "T has at most $2^{\lambda-1} = 2^{1-1} = 1$ node" is true, completing the basis step. The *recursive step* generates tree $T = T_1 \cdot T_2$. The number of levels in T is $k+1$ where k is the larger of the number of levels in either T_1 or T_2. Since we can assume the inductive hypothesis holds for the subtrees T_1 and T_2, we have that the number of leafs in T_1 is at most 2^{k-1} and the number of leafs in T_2 is also at most 2^{k-1}. Because all the leaves of T must be leaves of T_1 or of T_2, the number of leaves in T is at most $2^{k-1} + 2^{k-1}$, which is 2^k. This concludes the recursive step. Thus part (a) is true by the principle of structural induction.

8.3 *Hint 1:* Each of these statements can be proved using structural induction. Proof outlines are similar to the proofs in Examples 39 and 40.